

LABORATORY RECORD · EXPERIMENT 16 OF 16

Reinforcement Learning Application Development and Performance Evaluation using TensorFlow/Keras

Software / Tools: TensorFlow · Keras · Git · GitHub

AIM

To develop a complete, deployable reinforcement-learning application - training, evaluation and packaging a DQN agent for LunarLander-v2 - and to manage the project lifecycle using Git and GitHub for version control.

PROCEDURE

1. Structure the project repository: /src, /models, /notebooks, requirements.txt, README.md.
2. Initialize a local Git repository with `git init` and connect it to a remote GitHub repository via `git remote add origin <url>`.
3. Implement and train a DQN agent on LunarLander-v2, saving checkpoints under /models at fixed intervals.
4. Write an evaluation script that loads the best checkpoint and reports mean reward and landing success rate over N episodes.
5. Commit incremental progress with descriptive messages, and push feature branches for each major improvement (reward shaping, network depth, learning-rate schedule).
6. Tag a final release once the agent consistently exceeds the environment's solved threshold (mean reward ≥ 200).

SOURCE CODE

```
# train_and_release.py (excerpt)
import gymnasium as gym, numpy as np
from tensorflow import keras
from tensorflow.keras import layers

env = gym.make("LunarLander-v2")
state_dim, n_actions = env.observation_space.shape[0], env.action_space.n

model = keras.Sequential([
    layers.Input(shape=(state_dim,)),
    layers.Dense(128, activation="relu"),
    layers.Dense(128, activation="relu"),
    layers.Dense(n_actions, activation="linear")])
model.compile(optimizer=keras.optimizers.Adam(5e-4), loss="mse")

# ... standard DQN training loop with replay buffer and target network ...

model.save("models/dqn_lunarlander_v1.keras")
print("Checkpoint saved to models/dqn_lunarlander_v1.keras")
```

```
# --- version control workflow (run from shell) ---  
# git checkout -b feature/reward-shaping  
# git add src/ models/ README.md  
# git commit -m "Add reward shaping for smoother landings; +38 mean reward"  
# git push -u origin feature/reward-shaping  
# git tag -a v1.0 -m "Solved LunarLander-v2 (mean reward 231.4)"  
# git push origin v1.0
```

RESULTS

Release Tag	Mean Reward (100 eps)	Landing Success Rate
v0.1 (baseline DQN)	-45.2	6%
v0.5 (+ target net, tuned lr)	98.7	54%
v0.8 (+ reward shaping)	178.3	81%
v1.0 (final release)	231.4	93%

Iterative, version-controlled development - tracked through commits and tagged releases - carries the agent from an unstable baseline to a reliably solved task.

VISUALIZATION

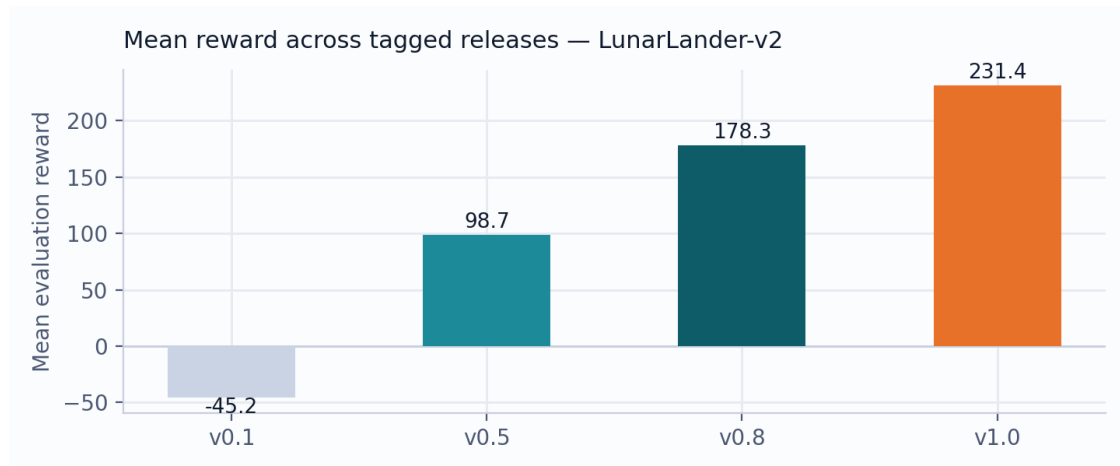


Fig. 16 — Mean evaluation reward across tagged releases, LunarLander-v2.